EXERCISE 4

FD6ACU

I. Choose the best answer.

1. The statistical device which helps in analyzing the co-variation of two or more variables is

   (a) variance (b) probability

   (c) correlation coefficient (d) coefficient of skewness

2. "The attempts to determine the degree of relationship between variables is correlation" is the definition given by

   (a) A.M. Tuttle (b) Ya-Kun-Chou

   (c) A.L. Bowley (d) Croxton and Cowden

3. If the two variables do not have linear relationship between them then they are said to have

   (a) positive correlation (b) negative correlation

   (c) uncorrelated (d) spurious correlation

4. If all the plotted points lie on a straight line falling from upper left hand corner to lower right hand corner then it is called

   (a) perfect positive correlation (b) perfect negative correlation

   (c) positive correlation (d) negative correlation

5. If $r = +1$, then the correlation is called

   (a) perfect positive correlation (b) perfect negative correlation

   (c) positive correlation (d) negative correlation

6. The correlation coefficient lies in the interval

   (a) $-1 \le r \le 0$ (b) $-1 < r < 1$ (c) $0 \le r \le 1$ (d) $-1 \le r \le 1$

7. Rank correlation coefficient is given by

   (a) $1+\frac{6\sum_{i=1}^{n} D_i^2}{n^3-n}$ (b) $1-\frac{6\sum_{i=1}^{n} D_i^2}{n^3-n}$ (c) $1-\frac{6\sum_{i=1}^{n} D_i^2}{n^3+n}$ (d) $1-\frac{6\sum_{i=1}^{n} D_i^3}{n\left(n^2-1\right)}$

8. If $\sum D^2 = 0$, rank correlation is

   (a) 0 (b) 1 (c) 0.5 (d) –1

9. Rank correlation was developed by

   (a) Pearson (b) Spearman (c) Yule (d) Fisher

10. Product moment coefficient of correlation is

    (a) $r=\frac{\sigma_x\ \sigma_y}{\operatorname{cov}(x,y)}$ (b) $r=\sqrt{\sigma_x\ \sigma_y}$ (c) $r=\frac{\operatorname{cov}(x,\ y)}{\sigma_x\ \sigma_y}$ (d) $r=\frac{\operatorname{cov}(x,\ y)}{\sigma_{xy}}$

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